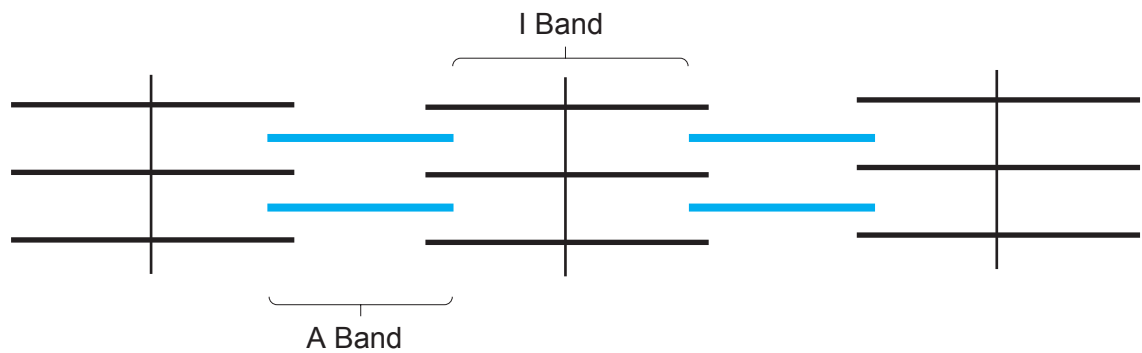


Option B: Human Musculoskeletal Anatomy

8. In 1953, Huxley introduced the sliding filament theory to explain muscle contraction. This theory was based on the idea that muscle proteins slide past each other to generate tension. Below is a diagram of two sarcomeres.



(a) (i) Identify the main proteins found within each of the regions using ticks (✓) to complete the following table. [2]

	Actin	Myosin	Troponin	Tropomyosin
A Band				
I Band				

(ii) Explain how the interaction between actin and myosin results in the contraction of muscle. Reference to the roles of troponin and tropomyosin are **not** required. [2]

.....

.....

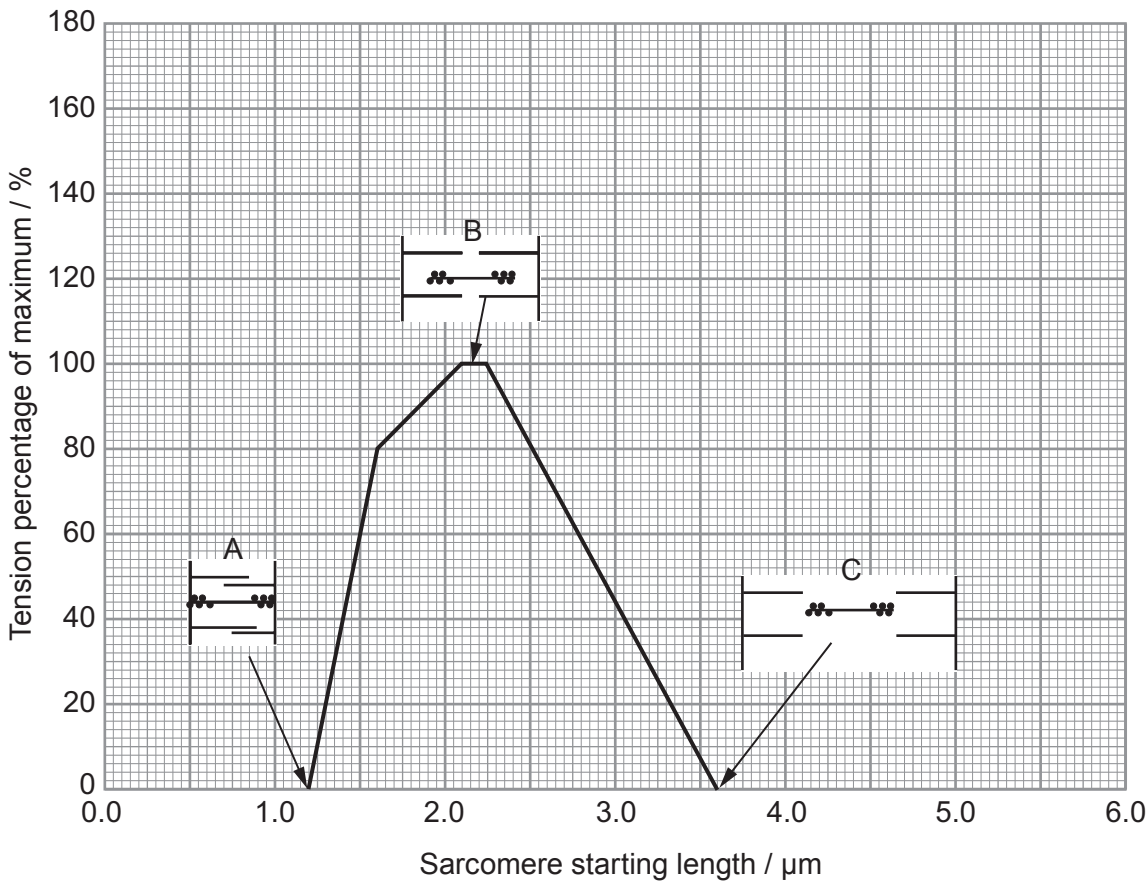
.....



Muscle samples can be analysed in order to produce a length-tension curve using the following procedure:

- muscle fibres are suspended in a solution
- muscles are positioned so that sarcomeres are at different lengths as shown at A, B and C on the graph below
- muscle fibres are stimulated to contract
- the tension (force) generated as a percentage of the maximum is measured at each sarcomere length (A, B and C)

The results are shown on the graph below (**diagrams of sarcomeres are not drawn to scale**).



- (b) (i) To determine the length of sarcomeres at different resting positions, myofibrils were viewed using a high-powered microscope. The muscle proteins were stained using fluorescent chemicals that bonded to the actin and myosin. Explain why at least 20 sarcomeres were measured. [1]

.....

.....



Examiner
only

(ii) Use the **graph** to calculate the length of the actin fibre. [3]

actin length =

(iii) Physiologists conducting these experiments used tissue from the same organism. Suggest **two** other factors that would need to be controlled when carrying out the investigation. [2]

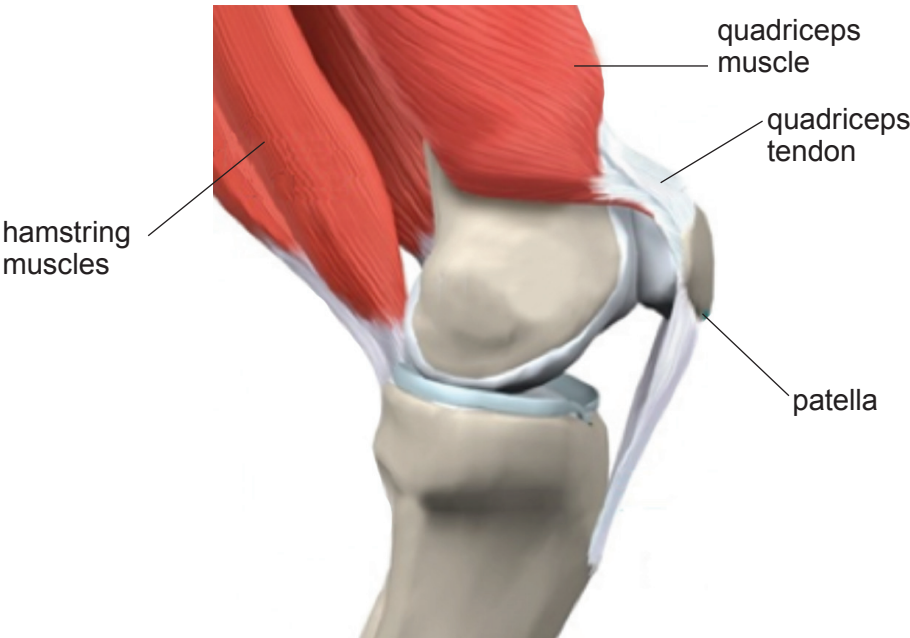
.....
.....

(iv) Suggest an explanation as to why no tension is generated at **C** on the graph. [2]

.....
.....
.....



- (c) Movement of the legs in humans involves the use of muscles attached by tendons to the bones. The quadriceps and hamstring muscles work antagonistically to move the knee joint and raise the leg. The image below shows the arrangement of muscles and tendons in the knee joint.



- (i) Explain why skeletal muscles are arranged in antagonistic muscle pairs. [1]

Scientists studied patients with osteoarthritis and a control group without the condition. Some of the measurements made are shown in the table below.

	Osteoarthritis Group	Control Group
Mean age	65	65
Maximum voluntary contraction force / N	444.50	486.10
Stair climbing time / s	13.08	8.82

- (ii) With reference to the role of the quadriceps in movement of the lower leg, explain the differences between the osteoarthritis and control groups. [2]

.....

.....

.....

